Pending Confirmation

If you book an appointment with status `02` (**Pending Confirmation**), the system tries to book an appointment for the time slot that is defined by the requested start date time and the requested end date time. If this succeeds, a booked appointment with status `03` is created. If it fails due to missing capacity, only an appointment request with status `01` is created.

Guest User Access

You can provide a guest user access to your customers and carriers who are not on SAP Business Network for Logistics and enable them to maintain the appointment details in SAP Business Network for Logistics system.

Once you provide a guest user access, the system generates an URL and sends it to the carrier in an email. To validate the authenticity of the accessing personnel, when the carrier navigates to the URL, the system prompts the carrier to enter additional appointment information such as the order number or delivery ID of the appointment. Once the carrier passes the security validation, the carrier can view the appointment details, select the time slots, and save the appointment. The customer can also invite a carrier to access this appointment and select the time slots.

The appointments which allow guest user access must contain the following:

- Additional customer contact details such as the email address
- Appointment type as EX
- Role code as `LBN_02`
- Document reference type as shown in the table under **Structure** for `DocumentType`.
- The standard IDs of the `SenderParty` and `RecipientParty` header fields must be set to your own LBN ID.

You must also retrieve the email address of the customer from the business partner directory in your external system and pass the information through this service.

Create an Appointment Without Reference Document

The system enables you to create an appointment without a reference document. You can later add the details such as reference document details, order details, delivery ID, and so on. You can also add additional reference document details or replace a reference document.

Enable Overbooking

When you enable overbooking for a loading point, you can create multiple appointments or block a time slot multiple times irrespective of the capacity of the loading point. You can also enable overbooking only for a carrier or only for you as a shipper depending on your settings for that loading point.

In the appointment confirmation, a new node has been added for messages. You can view the messages in the following scenarios:

- If a customer creates an appointment via API, the system creates a confirmation message for the same.
- If the loading point is not set for overbooking and if preferred time slot is not available, the system does not create the appointment and creates an error message for the same.
- When the customer sends the appointment with the status as **Pending Confirmation** and the system changes the appointment status to either **Confirmed** or **Appointment Request Created**, the system provides a message for the same.

Preferred Slots / Slots Preferred by Shipper

When you maintain an appointment, the system displays the requested and acceptable dates for slot booking based on the following criteria:

- The requested dates are taken from the `RequestedStartDateTime` and `RequestedEndDateTime` fields.
- The acceptable dates are taken from the `AcceptableStartDateTime` and `AcceptableEndDateTime` fields.

Notes and Attachments

You can use this inbound API to update the SAP Business Network for Logistics system with the notes and attachments from your external systems.

The attachment name must contain the file extension. Example: *Attachment.pdf*.

Automatic Cancellation of Appointments

An appointment is automatically canceled if one of the following is changed:

- Loading point
- Docking location
- Acceptable dates (start or end date)
- Requested dates (start or end date)

The following exceptions apply:

- If the changed requested dates match the start and end date of the scheduled appointment, the appointment will not be canceled.
- If the start and end date of the scheduled appointment is within the range of the new acceptable start and end date, the appointment will also not be canceled.

Provider API for Dock Appointment Cancellation

As a shipper, you can use this inbound service to cancel the dock appointments in SAP Business Network for Logistics based on the information from your external system. You can get the details from your external systems such as a transportation system, warehouse management system, yard management system, and so on.

> **Note**
>
> To avoid messages out of sequence or late runners overwriting newer updates, the processing of these late runner messages will be skipped.
>
> The `MessageHeader` → `CreationDateTime` field will be used to determine if the update will be processed or skipped. In case the `CreationDateTime` will not be present, the message will always be processed as the latest update.

Technical Name

`GenericAppointmentCancellation_In`

Prerequisites

You have performed the following:

- You have created a shipper and carrier business partner in your SAP Business Network for Logistics and your external system.
- In your external system, you have the LBN ID and you have mapped the LBN ID with the business partner.

Details

To cancel an appointment in SAP Business Network for Logistics, you must send the appointment ID of the appointment created or updated in your SAP Business Network for Logistics system from the external system.

Consumer API for Dock Appointment Scheduling

As a shipper, you can use this outbound service provided by SAP Business Network for Logistics to update your external systems with the appointment details. You can update your external systems such as a transportation system, warehouse management system, yard management system, and so on with the appointment details.

Technical Name

`GenericAppointment_Out`

Description

You can update a freight order, transportation unit, or yard order with the appointment ID and the appointment details created in your SAP Business Network for Logistics.

Structure

The system has different fields categorized under different nodes. Each node contains various fields. Below are some of the important fields displayed by the system:

Technical Name	Field	Description	Node
OrderingPartyLbnId	Ordering Party LBN ID*	Contains the ordering party ID maintained in the SAP Business Network for Logistics system	Appointments
CarrierLbnId	Carrier LBN ID*	Contains the carrier ID maintained in the SAP Business Network for Logistics system	Appointments
SenderSystemId	Sender System ID*	Contains the ID of sender system	Appointments
AppointmentStatusCode	Appointment Status Code*	Passes one of the following status codes: 01 – Appointment Request Created 03 – Confirmed 04 – Rejected	Appointments

Technical Name	Field	Description	Node
		05 – Freight Order Canceled 06 – Canceled by Operator 07 – Canceled by Carrier 09 - Canceled by Generic Appointment API	
LoadingActivityType	Loading Activity Type	Contains one of the following activity types: 01 – Loading 02 – Unloading 03 – Loading/Unloading	Appointments
Text	Text	Contains the reason why the appointment is canceled by the carrier or warehouse operator.	Appointment Note
CreatedAt	Created At	Contains the date and time at which the note was created	Appointment Note
DocumentId	Document ID	Contains the document ID of the source business document from your external system (For example, freight order, transport unit, yard order).	Appointment Reference Document
DocumentType	Document Type	Contains one of the following types of reference document: 1311 –Transportation Unit Appointment (Extended Warehouse Management) 1122 –Transportation Order (Transportation Management) YO –Yard Order (Yard Logistics)	Appointment Reference Document

Note that the fields marked with * are mandatory fields.

ℹ Note

As of now, we do not support the extensibility feature with this service.

Notes and Attachments

You can use this outbound API to update the external systems with the notes from your SAP Business Network for Logistics.

You can use this outbound API to update the external systems with the attachments from your SAP Business Network for Logistics.

Share Appointments and Appointment Requests with Business Partners

You can share a docking location with your business partners in SAP Business Network for Logistics. When an appointment request is received or an appointment is booked with this docking location in the SAP Business Network for Logistics system, you can use this outbound API to update the appointment details in the external systems. Along with your external systems, the system also updates the appointment details in the external systems of business partners with whom you have shared this docking location.

Receive Error Messages in your External System

You can use this API to receive error messages in your external system in case of business validation errors that may occur while processing your request to SAP Business Network for Logistics. This API sends the following information:

- Processing status

  Displays whether the status is blank (success), 1 (info), 2 (warning), or 3 (error)

- Correlation ID

  You can use this ID to view or reprocess the message in **Manage Integration Logs** app

- Log node

  Additional node in the API that contains the message code, message, and the severity code

When a business validation error occurs, the system fills the correlation ID in the message and sets the processing status as 3 (Error). This indicates that the message processing has failed. You can use this correlation ID to find this message in the **Manage Integration Logs** app and check for the issue. The system also fills the log node describing the issue that occurred while processing the request.

You can view below the severity codes and the error messages that can occur.

Severity Codes

- Blank (success)
- 1 (info)
- 2 (warning)
- 3 (error)

Message Codes and Message

Message Code	Message
10001	The appointment type code is invalid
10003	The carrier/ordering party LBN ID is invalid
10004	The docking location is invalid
10005	The status of the appointment is invalid
10006	The appointment date/time is invalid
10007	The docking location or loading point is invalid
10008	The appointment duration is invalid
10009	The selected date is a non-working day.

Message Code	Message
10010	No time slots are available. Please contact the warehouse operator to check docking location settings.
10011	The time slot you are trying to book does not fit any period specified in business hours
10012	No time slots are available. The selected date is fully booked.
10013	Appointment does not exist
10014	Found multiple existing appointments

Provider API to Manage Gate Operations

As a shipper, you can use this inbound service to update the gate logbook details in your SAP Business Network for Logistics system based on the vehicle check-in and checkout information from your external systems such as a transportation system, warehouse management system, yard management system and so on.

ℹ Note

To avoid messages out of sequence or late runners overwriting newer updates, the processing of these late runner messages will be skipped.

The `MessageHeader` → `CreationDateTime` field will be used to determine if the update will be processed or skipped. In case the `CreationDateTime` will not be present, the message will always be processed as the latest update.

Technical Name

`GenericGateProcess_In`

Prerequisites

- The appointment has been confirmed.
- The entity settings configuration is done, this is, the loading point, docking location, gate and yard have been created.
- An appointment ID exists.
- Gate and yard have been assigned to the location.

Description

You can perform the following using this API:

- **Check-In Process**

  If the carrier confirms an appointment, the system creates a gate logbook in the SAP Business Network for Logistics system and notifies the external systems using the `GenericGateProcess_Out` API. The gate operations manager can directly begin the check-in process in the external system.

  The gate operations manager updates the check-in details such as the check-in time, driver details, gate details, vehicle licence plate, weight and seal details, equipment details, notes, and attachments in the external system. You can use this inbound API to update your SAP Business Network for Logistics system with these details.

- **Checkout Process**

  Once the vehicle checks out from the gate, the gate operations manager updates the checkout details such as the checkout time, driver details, gate details, vehicle license plate, weight and seal details, equipment details, notes, and attachments in the external system. When you specify a seal number, you can enter an alphanumeric value. Note that you must confirm the driver details before you can proceed with the checkout of the vehicle. You can use this inbound API to update your SAP Business Network for Logistics system with these details.

- **Request Documents from the External System**

  You can use this inbound API to send additional documents such as loading list and bill of lading from your external transportation system to your SAP Business Network for Logistics system.

ℹ Note

The network accepts only 700 cargo items in an API call.

You can send one or more of the following statuses as applicable, during the gate operation process:

Status	Status Code	Description
`READY_FOR_CHECKIN`	01	Send this status when the gate logbook is created. You can update the gate logbook before the check-in begins, for example to update documents.
`CHECKIN_IN_PROGRESS`	02	Send this status when the gate operations manager starts the check-in process
`CHECKIN_ON_HOLD`	03	Send this status when the gate operations manager puts the check-in process on hold
`CHECKIN_SUCCESSFUL`	04	Send this status when the gate operations manager completes the check-in process
`CHECKOUT_IN_PROGRESS`	06	Send this status when the gate operations manager starts the checkout process
`CHECKOUT_ON_HOLD`	07	Send this status when the gate operations manager puts the checkout process on hold
`CHECKOUT_SUCCESSFUL`	08	Send this status when the gate operations manager completes the checkout process
`CHECKIN_CANCELED`	09	Send this status to cancel the check-in process

Reporting Events

You can use this API to update your SAP Business Network for Logistics system with the events reported from your external systems.

Using this API, during the check-in and checkout processes at a gate, the system updates the gate logbook document in SAP Business Network for Logistics system with the events reported by one or more business partners.

You must send the following parameters in this API:

Technical Name	Field	Description
eventType	Event Type	Contains one of the following event types: • GATEIN_START – Start check-in • GATEIN_END – Check-in complete • GATEOUT_START – Start checkout • GATEOUT_END – Checkout complete • DELAYED - Delayed
eventReasonCode	Event Reason Code	Contains one of the event reason codes for the event type DELAYED given in the table below.
eventReasonDescription	Event Reason Description	Contains the respective description of the event reason code as given in the table below.
actualBusinessTimestamp	Actual Event Time	Contains the actual event time
actualBusinessTimezone	Actual Event Time Zone	Contains the time zone of the actual event

DELAYED event reason codes and descriptions:

Event Reason Code	Description
LBN_TRAFFC	Traffic
LBN_BRKDWN	Breakdown
LBN_WTGLOC	Waiting at Location
LBN_OTHERS	Others
CANCELLATION	Cancellation
CLEANING_STATION	Cleaning Station
CLEANLINESS_NOT_FIT	Cleanliness not fit
COMM_INSTRUCTION	Communications/Instructions
CONGESIONTOFBAYS	Congestion of Bays
CROSSOVER	Crossover
DOCUMENTMISSING	Documents Missing/IT Failure
ACCIDENT	Equipment Accident or Malfunction
FORCEOFNATURE	Force of Nature
INCORRECT	Incorrect or Not Conform
LACKOFSTORAGECAPA	Lack of Storage Capacity
LOADINGSECURING	Load Securing

Event Reason Code	Description
NEARMISS	Near Miss
NOTAVAILABLE	Not Available
PACKAGINDAMAGED	Packaging Damaged
PREVIOUSLOADINGORUNLOADING	Previous loading or unloading Point
DAMAGED	Product Damaged/ Contaminated
PUBLICAUTHORITY	Public Authorities
QUANTITY	Quantity
SEALCOMPROMISED	Seal Compromised
SPILL	Spill
STRIKE	Strike
SURVEYOR	Surveyor
TEMPERATURE	Temperature
TERMINAL	Terminal/ Rail/ Shipping Line
THEFT	Theft/Vandalism
UNSAFE	Unsafe Condition
WRONGBEHAVIOU	Wrong Behaviour

Consumer API to Manage Gate Operations

You can use this outbound service to update your external systems with the check-in and checkout details provided by SAP Business Network for Logistics. You can update your external systems such as a transportation system, warehouse management system, yard management system, and so on with the check-in and checkout details.

Technical Name

`GenericGateProcess_Out`

Description

You can perform the following using this API:

- **Send Check-In Details to your External System**

  Once the SAP Business Network for Logistics system is updated with the check-in details, the system uses this outbound API to send the check-in details from your SAP Business Network for Logistics system to your external systems. The system also sends the details to the external systems of all the business partners who are associated with this gate logbook.

- **Send Weight Details to your External System**

You can use this outbound API to send weight details from your SAP Business Network for Logistics system to your external systems.

- **Send Documents to your External System**

  You can use this outbound API to send reference documents such as loading list, bill of lading from your SAP Business Network for Logistics system to your external systems.

- **Send the Checkout Details to your External System**

  Once the SAP Business Network for Logistics is updated with the checkout details, the system uses this API to send the checkout details from your SAP Business Network for Logistics system to your external systems.

The system sends one or more of the following statuses as applicable, during this process:

Status	Status Code	Description
READY_FOR_CHECKIN	01	Sends this status when the gate logbook is created
CHECKIN_IN_PROGRESS	02	Sends this status when the gate operations manager starts the check-in process
CHECKIN_ON_HOLD	03	Sends this status when the gate operations manager puts the check-in process on hold
CHECKIN_SUCCESSFUL	04	Sends this status when the gate operations manager completes the check-in process
CHECKOUT_IN_PROGRESS	06	Sends this status when the gate operations manager starts the checkout process
CHECKOUT_ON_HOLD	07	Sends this status when the gate operations manager puts the checkout process on hold
CHECKOUT_SUCCESSFUL	08	Sends this status when the gate operations manager completes the checkout process

Receive Error Messages in your External System

You can use this API to receive error messages in your external system in case of business validation errors that may occur while processing your request to SAP Business Network for Logistics system. This API sends the following information:

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- Correlation ID

  You can use this ID to view or reprocess the message in **Manage Integration Logs** app

- Log node

  Additional node in the API that contains the message code, message, and the severity code

When a business validation error occurs, the system fills the correlation ID in the message and sets the processing status as 3 (Error). This indicates that the message processing has failed. You can use this correlation ID to find this message in the **Manage Integration Logs** app and check for the issue. The system also fills the log node describing the issue that occurred while processing the request.